IVS

2026-06-11

Packages

```r
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```r
library(arrow)
```

```
## Warning: package 'arrow' was built under R version 4.5.3
```

```
##
## Attaching package: 'arrow'
```

```
## The following object is masked from 'package:utils':
##
##     timestamp
```

```r
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(tidyr)
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.3
```

```
library(DataExplorer)
```

```
## Warning: package 'DataExplorer' was built under R version 4.5.3
```

```
library(knitr)
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

Dataset

```
ivs <- read_parquet("C:/Users/alyss/Documents/GitHub/intox_agro/resultados/contextual/ivs_munici
pios_sp_2010.parquet")
```

Cleaning

```
#filter
ivs_clean <- ivs |>
  filter(label_sit_dom == "Total Situação de Domicílio")

ivs_clean <- ivs_clean |>
  filter(label_sexo == "Total Sexo")

ivs_clean <- ivs_clean |>
  filter(label_cor == "Total Cor")
```

```
#variable class

class(ivs_clean$ivs_renda_e_trabalho)
```

```
## [1] "character"
```

```
class(ivs_clean$renda_per_capita)
```

```
## [1] "character"
```

```
class(ivs_clean$i_gini)
```

```
## [1] "character"
```

```
class(ivs_clean$t_analf_15m)
```

```
## [1] "character"
```

```
class(ivs_clean$ivs_capital_humano)
```

```
## [1] "character"
```

```
class(ivs_clean$t_sem_agua_esgoto)
```

```
## [1] "character"
```

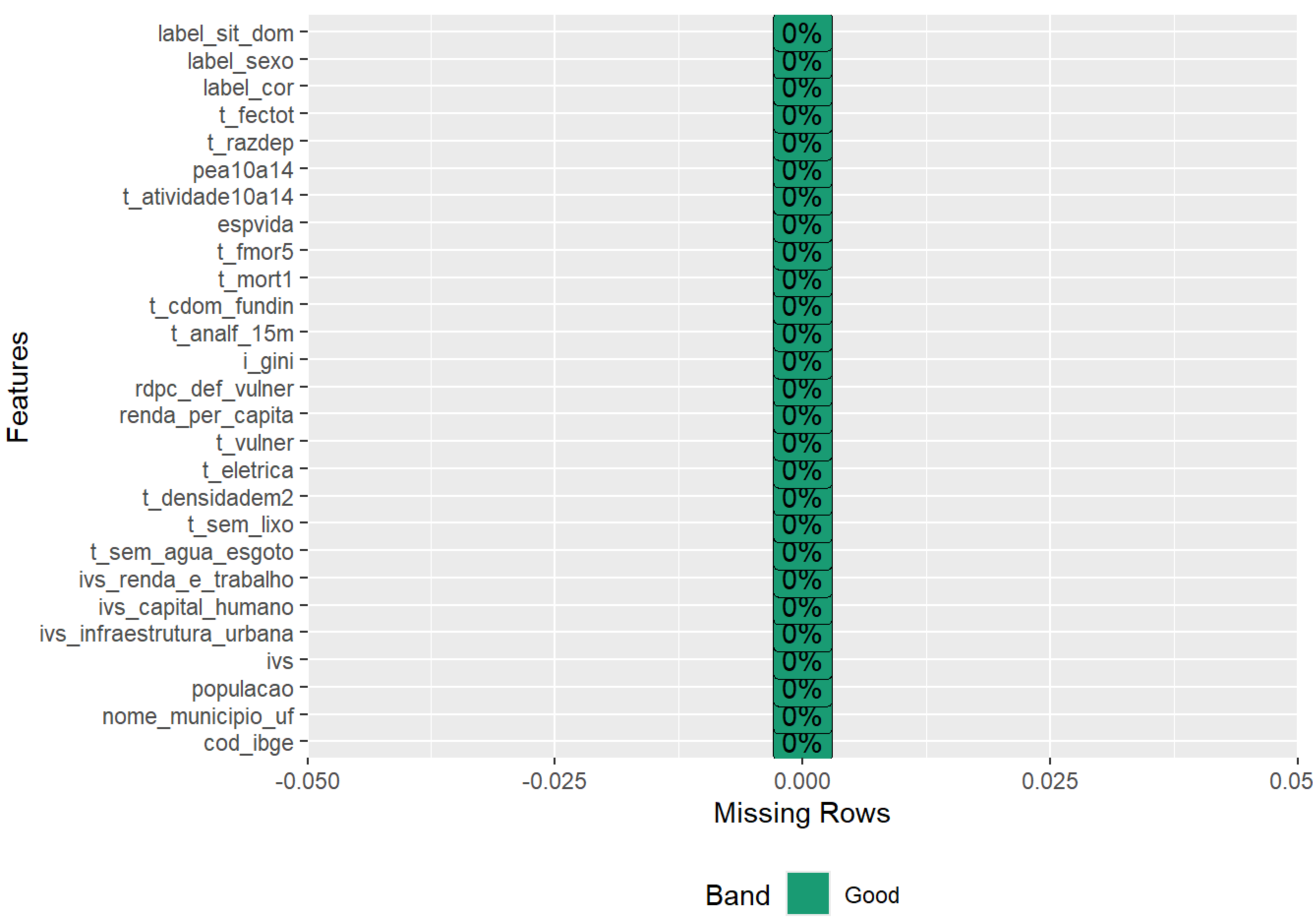
```
class(ivs_clean$t_sem_lixo)
```

```
## [1] "character"
```

```
#make '-' NA

ivs_clean[ivs_clean == "-"] <- NA

#make numeric

ivs_clean <- ivs_clean %>%
  mutate(
    across(
      c(
        ivs_renda_e_trabalho,
        renda_per_capita,
        i_gini,
        t_analf_15m,
        ivs_capital_humano,
        t_sem_agua_esgoto,
        t_sem_lixo
      ),
      ~ as.numeric(gsub(",", ".", .))))
```

Missingness

```
plot_missing(ivs_clean)
```



Income & Employment

Income and Labor Sub-Index

Distribution

```
ivs_renda_e_trabalho_summary <- ivs_clean %>%
  select(ivs_renda_e_trabalho) %>%
  rename(value = ivs_renda_e_trabalho) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Income and Labor Sub-Index",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
ivs_renda_e_trabalho_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Income and Labor Sub-Index",
    col.names = c(
      "Income and Labor Sub-Index",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

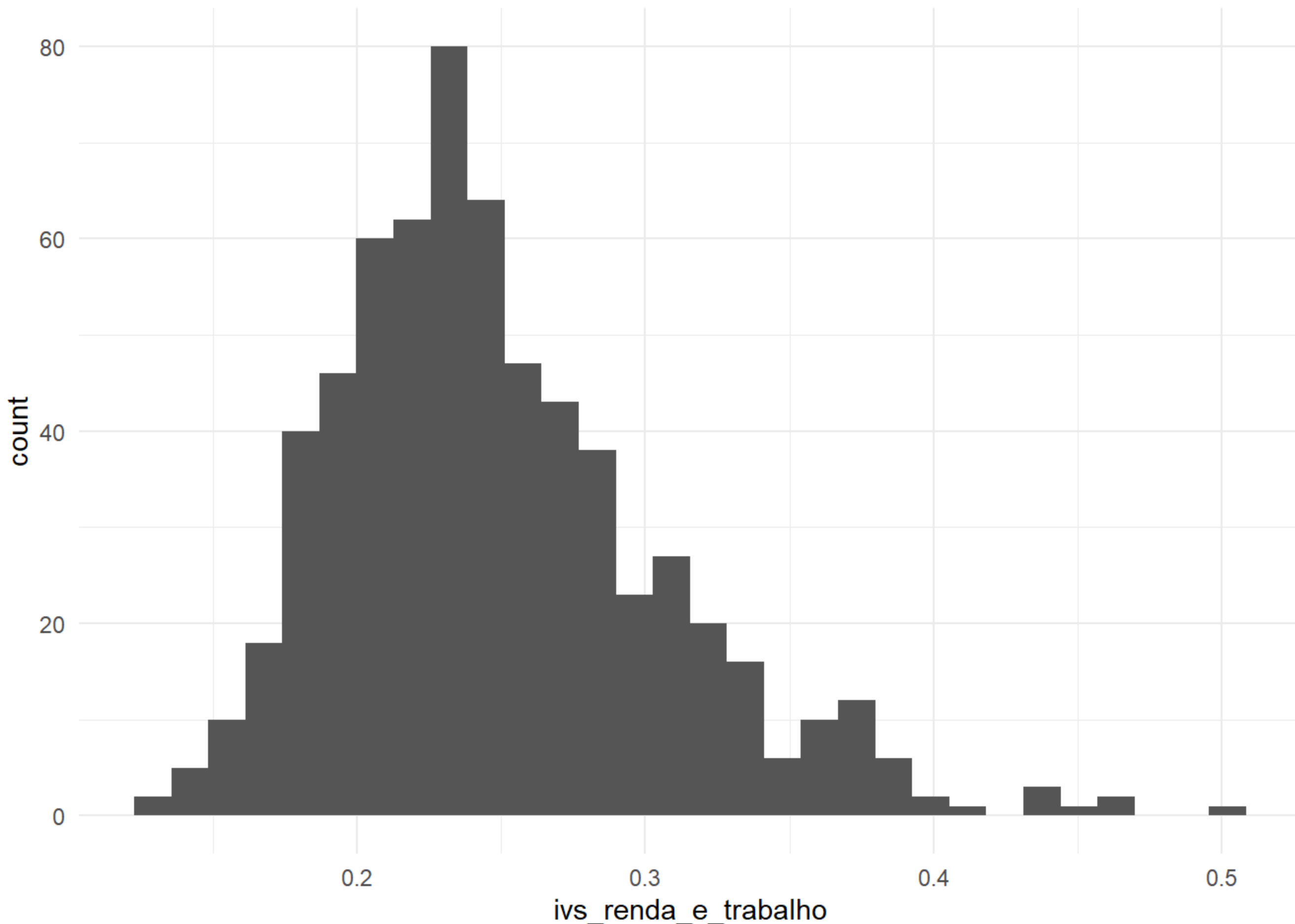
Distribution of Municipalities by Income and Labor Sub-Index

Income and Labor Sub-Index	Quartile	n	%	Median (%)	Range

Income and Labor Sub-Index	Quartile	n	%	Median (%)	Range
Income and Labor Sub-Index	1	162	25.1	0.3	0.28-0.5
	2	161	25.0	0.3	0.24-0.28
	3	161	25.0	0.2	0.21-0.24
	4	161	25.0	0.2	0.13-0.21

Normality

```
ggplot(ivs_clean, aes(x = ivs_renda_e_trabalho)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Per-Capita Income

Distribution

```
renda_per_capita_summary <- ivs_clean %>%
  select(renda_per_capita) %>%
  rename(value = renda_per_capita) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Per-Capita Income",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
renda_per_capita_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Per-Capita Income",
    col.names = c(
      "Per-Capita Income",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

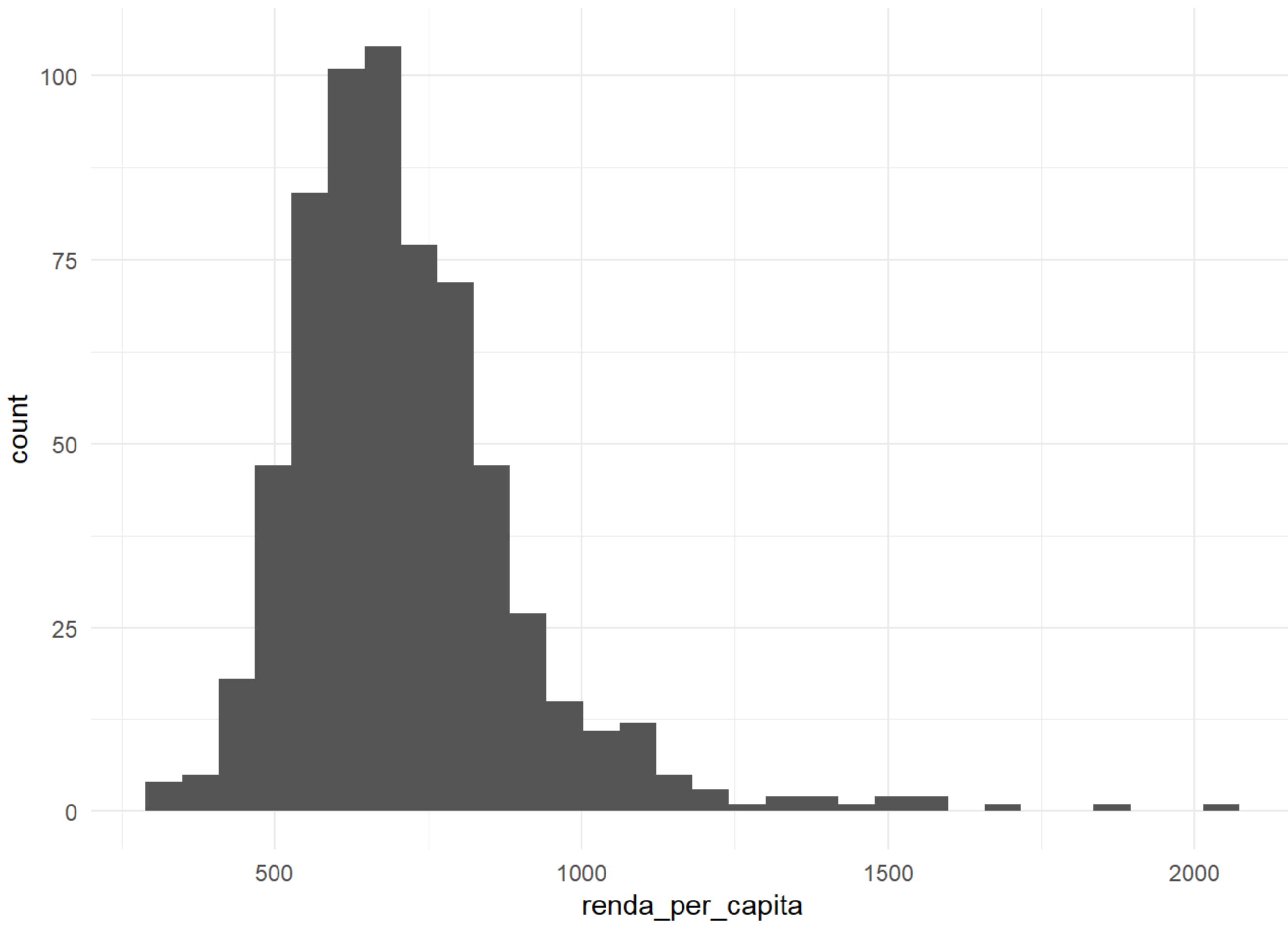
Distribution of Municipalities by Per-Capita Income

Per-Capita Income	Quartile	n	%	Median (%)	Range

Per-Capita Income	Quartile	n	%	Median (%)	Range
Per-Capita Income	1	162	25.1	906.0	798.74-2043.74
	2	161	25.0	731.5	686.89-798
	3	161	25.0	632.4	588.36-686.56
	4	161	25.0	533.7	318.44-587.83

Normality

```
ggplot(ivs_clean, aes(x = renda_per_capita)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Income Inequality - Gini Coefficient

Distribution

```
i_gini_summary <- ivs_clean %>%
  select(i_gini) %>%
  rename(value = i_gini) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Gini Index",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
i_gini_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Gini Index",
    col.names = c(
      "Gini Index",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

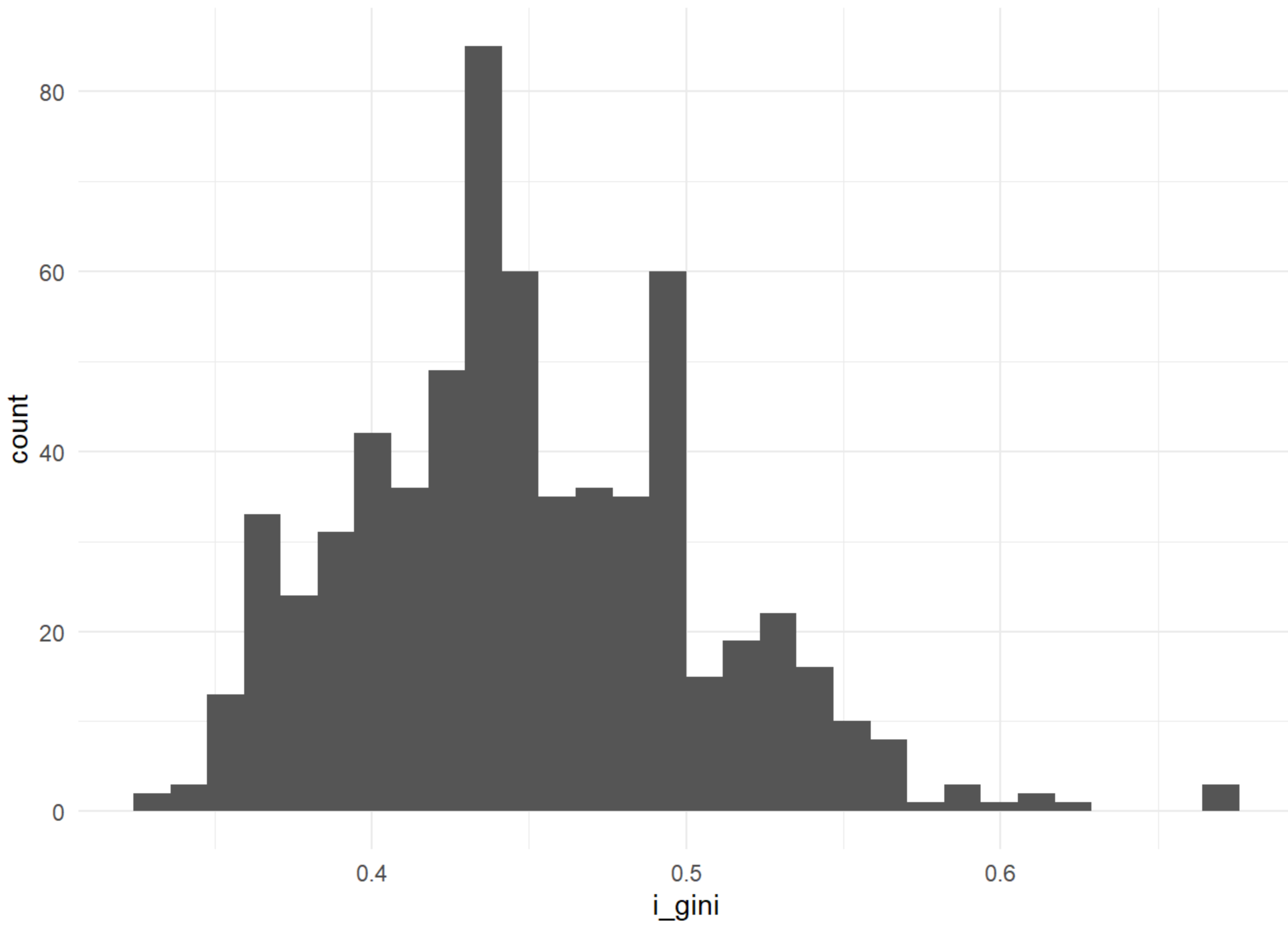
Distribution of Municipalities by Gini Index

Gini Index	Quartile	n	%	Median (%)	Range

Gini Index	Quartile	n	%	Median (%)	Range
Gini Index	1	162	25.1	0.5	0.48-0.67
	2	161	25.0	0.5	0.45-0.48
	3	161	25.0	0.4	0.41-0.45
	4	161	25.0	0.4	0.33-0.41

Normality

```
ggplot(ivs_clean, aes(x = i_gini)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Education

Adult Literacy Rate = > 15 years

Distribution

```
t_analf_15m_summary <- ivs_clean %>%
  select(t_analf_15m) %>%
  rename(value = t_analf_15m) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Adult Literacy Rate",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
t_analf_15m_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Adult Literacy Rate",
    col.names = c(
      "Adult Literacy Rate",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

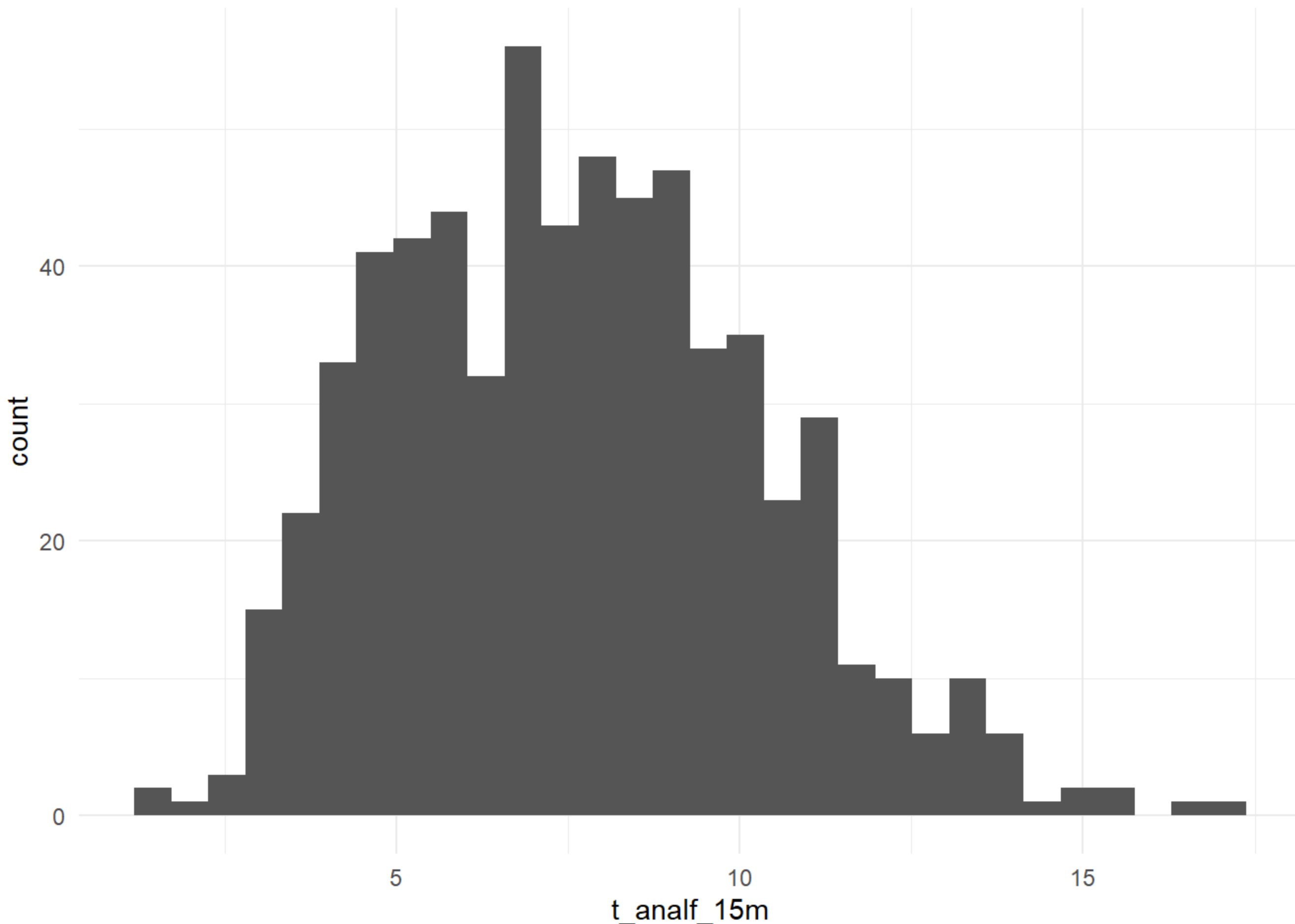
Distribution of Municipalities by Adult Literacy Rate

Adult Literacy Rate	Quartile	n	%	Median (%)	Range

Adult Literacy Rate	Quartile	n	%	Median (%)	Range
Adult Literacy Rate	1	162	25.1	10.9	9.46-17.1
	2	161	25.0	8.4	7.5-9.46
	3	161	25.0	6.6	5.56-7.49
	4	161	25.0	4.5	1.45-5.55

Normality

```
ggplot(ivs_clean, aes(x = t_analf_15m)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Human Capital Sub-Index

Distribution

```
ivs_capital_humano_summary <- ivs_clean %>%
  select(ivs_capital_humano) %>%
  rename(value = ivs_capital_humano) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Human Capital Sub-Index",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
ivs_capital_humano_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Human Capital Sub-Index",
    col.names = c(
      "Human Capital Sub-Index",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

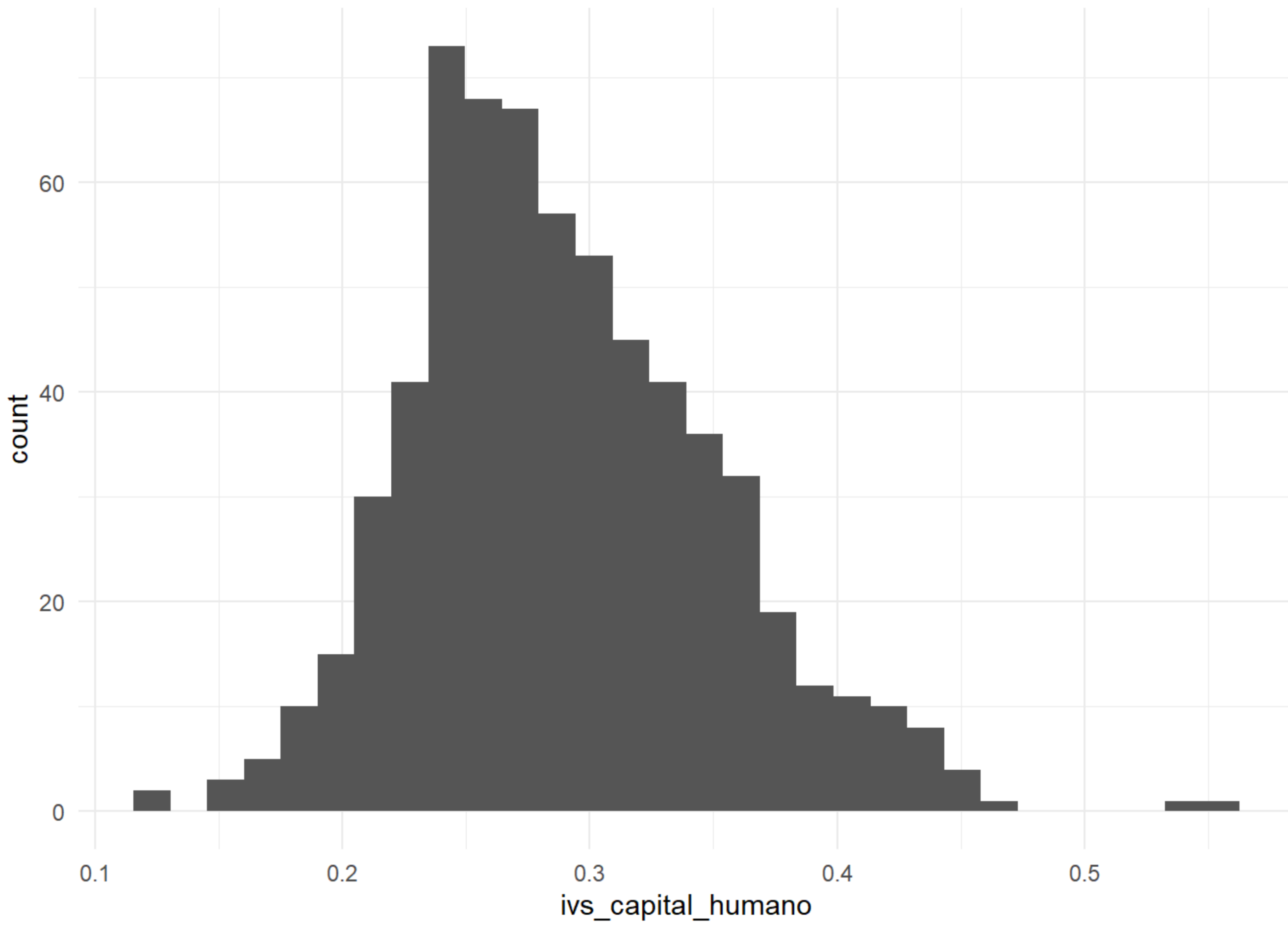
Distribution of Municipalities by Human Capital Sub-Index

Human Capital Sub-Index	Quartile	n	%	Median (%)	Range

Human Capital Sub-Index	Quartile	n	%	Median (%)	Range
Human Capital Sub-Index	1	162	25.1	0.4	0.33-0.56
	2	161	25.0	0.3	0.28-0.33
	3	161	25.0	0.3	0.24-0.28
	4	161	25.0	0.2	0.12-0.24

Normality

```
ggplot(ivs_clean, aes(x = ivs_capital_humano)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Sanitation & Environmental Exposure

Households without Piped Water or Sewage

Distribution

```
t_sem_agua_esgoto_summary <- ivs_clean %>%
  select(t_sem_agua_esgoto) %>%
  rename(value = t_sem_agua_esgoto) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Households without Piped Water or Sewage",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
t_sem_agua_esgoto_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Households without Piped Water or Sewage",
    col.names = c(
      "Households without Piped Water or Sewage",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

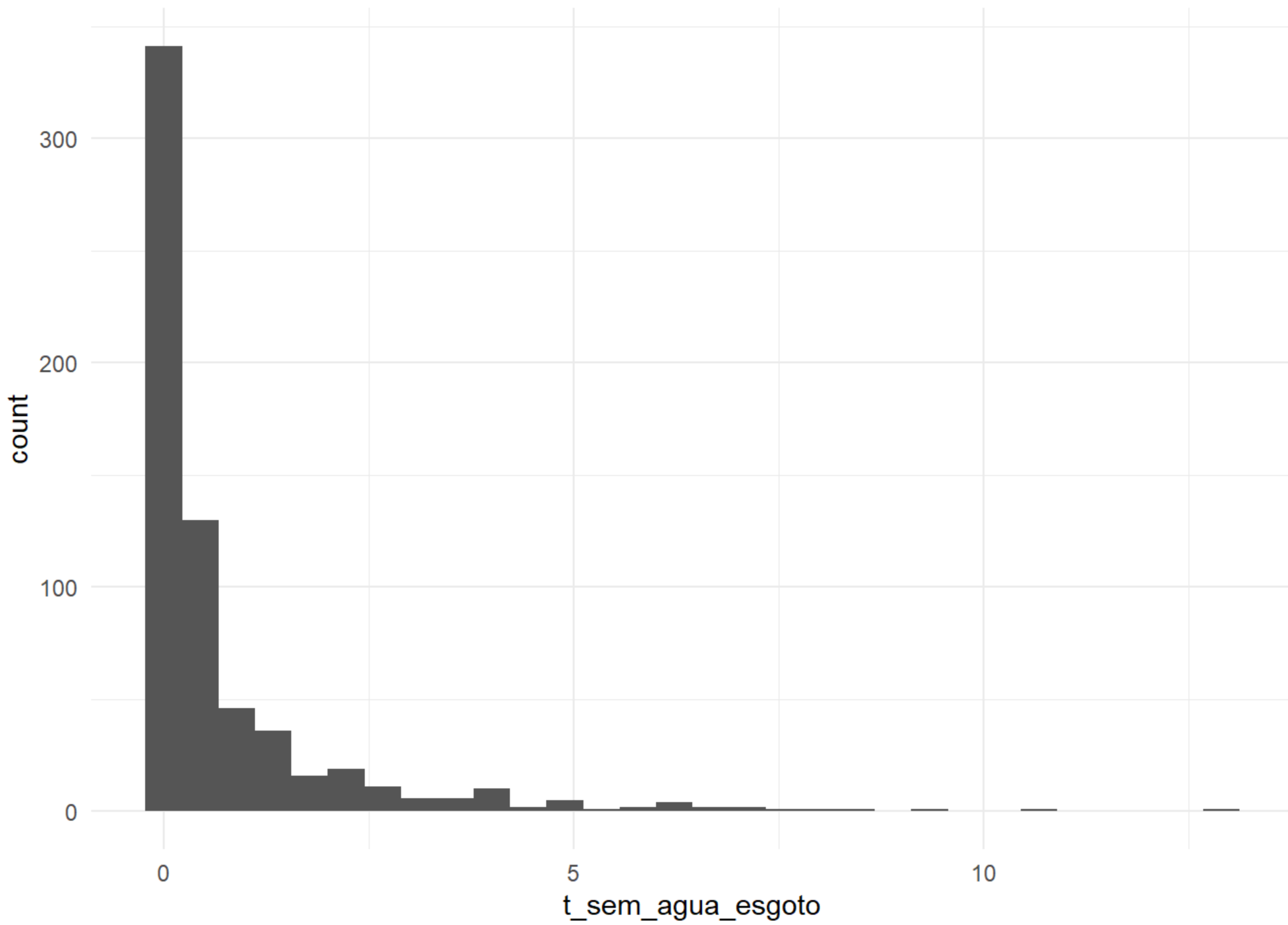
Distribution of Municipalities by Households without Piped Water or Sewage

Households without Piped Water or Sewage	Quartile	n	%	Median (%)	Range

Households without Piped Water or Sewage	Quartile	n	%	Median (%)	Range
Households without Piped Water or Sewage	1	162	25.1	1.9	0.77-12.9
	2	161	25.0	0.4	0.19-0.76
	3	161	25.0	0.1	0-0.19
	4	161	25.0	0.0	0-0

Normality

```
ggplot(ivs_clean, aes(x = t_sem_agua_esgoto)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```



Households without Garbage Collection

Distribution

```
t_sem_lixo_summary <- ivs_clean %>%
  select(t_sem_lixo) %>%
  rename(value = t_sem_lixo) %>%
  mutate(
    Quartile = ntile(desc(value), 4)
  ) %>%
  group_by(Quartile) %>%
  summarise(
    n = n(),
    pct = round(100 * n() / nrow(ivs_clean), 1),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Variable = "Households without Garbage Collection",
    Range = paste0(round(Min, 2), "-", round(Max, 2))
  ) %>%
  select(Variable, Quartile, n, pct, Median, Range)

#kable
t_sem_lixo_summary %>%
  kable(
    digits = 1,
    caption = "Distribution of Municipalities by Households without Garbage Collection",
    col.names = c(
      "Households without Garbage Collection",
      "Quartile",
      "n",
      "%",
      "Median (%)",
      "Range"
    )
  ) %>%
  collapse_rows(columns = 1) %>%
  kable_styling(full_width = FALSE)
```

Distribution of Municipalities by Households without Garbage Collection

Households without Garbage Collection	Quartile	n	%	Median (%)	Range

Households without Garbage Collection	Quartile	n	%	Median (%)	Range
Households without Garbage Collection	1	162	25.1	0.9	0.55-9.65
	2	161	25.0	0.3	0.24-0.55
	3	161	25.0	0.1	0-0.24
	4	161	25.0	0.0	0-0

Normality

```
ggplot(ivs_clean, aes(x = t_sem_lixo)) +
  geom_histogram(bins = 30) +
  theme_minimal()
```